

Part1 — Collect data and summarize how AI innovations and AI competitiveness are measured

To comprehensively and accurately diagnose Canada's global AI competitiveness, this study did not rely on a single indicator or data source but instead constructed a multi-source comprehensive database covering multiple core dimensions. We integrated open-source datasets from Kaggle, authoritative industry report websites, and strategic background documents provided by the course to form a unified analytical framework.

First, data from the initial domain constitutes the foundation for assessing national AI strength, sourced from *Kaggle (AI Global Index)* and *Tortoise Media*. Each country has an overall 0–100 score that summarizes its AI performance, along with nine underlying dimensions: talent, infrastructure, operating environment, research, development, government strategy, commercial strength, scale of activity, and intensity relative to country size.

Second, for Innovation Output & Quality, data were sourced from the *CB Insights* to extract Global Unicorn companies. To distinguish between "quantity" and "quality," we did not stop at counting the total number of unicorns across all industries (**Unicorn_Total_All_Industries**). Instead, we constructed a Core AI Unicorn Proxy (**Unicorn_AI_Smart_Proxy**) and AI Concentration (**AI_Concentration_Pct**), aiming to filter out non-technical enterprises and accurately measure a country's true output in the AI technology sector.

Third, the Society & Labor Force domain fills the blind spots of traditional macro indicators by focusing on the "human" factor, with data sourced from the *KPMG 2025 Report* and *Kaggle (Popularity Index)*. We introduced AI Literacy Rank (**AI_Literacy_Rank**) and Training Percentage (**AI_Literacy_Percent**) to quantify the workforce's capacity to adopt new technologies. Additionally, we combined this with Global Search Popularity (**Popularity_Score**) to capture public sentiment and market attention toward AI.

Fourth, regarding Strategic Intervention & Capital Investment, the data originated from files in the Google Drive provided by the professor. To quantify capital investment and national strategy, we introduced the Investment Gap Multiplier (**Investment_Gap_Multiplier_vs_Canada**) and Policy Centralization Score (**Policy_Centralization_Score**). These indicators transform qualitative strategic texts into quantifiable numerical features, enabling us to directly compare AI governance models under different political systems.

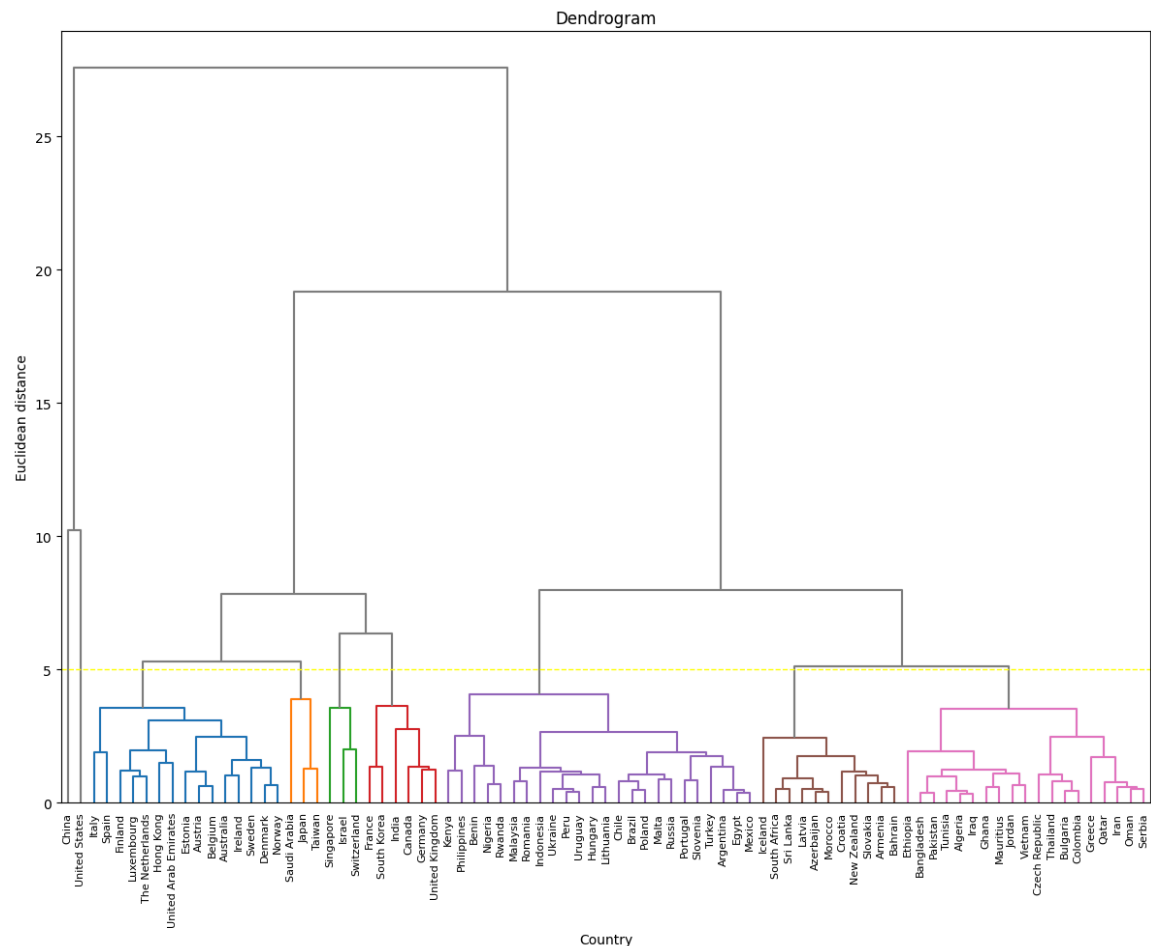
We use the country-level dataset introduced above to construct and interpret an AI innovation and competitiveness ranking. Talent, infrastructure, and operating environment are treated as indicators of AI competitiveness, because they capture the conditions that make AI adoption possible. Research and development are treated as indicators of AI innovation, while the remaining dimensions describe how innovation is commercialized and scaled.

Simple descriptive plots show that AI capabilities are extremely unevenly distributed. Histograms of the scores reveal that most countries lie between 0 and 30 on talent, research,

and development, while a small group of advanced economies extends into much higher values. This pattern appears across almost all dimensions: there is a long right tail containing countries such as the United States, China, the United Kingdom, Singapore, and a cluster of high-income states that have built both the human capital and the infrastructure needed for intensive AI activity.

Using the overall score, a global ranking of AI ecosystems is obtained. The United States is the clear leader with an overall score of 100, followed by China, Singapore, the United Kingdom, France, South Korea, Germany, Israel, Canada, and India. Canada’s overall score is 26, which places it around eighth in the world and firmly within the group of advanced AI countries, though still some distance from the scale and intensity of the top two.

To understand how countries resemble each other, hierarchical clustering is applied to the nine sub-scores (excluding the overall score), after standardizing them. The resulting dendrogram, with a dashed horizontal line at a distance of 5, reveals a small “power-player” cluster containing the United States and China, a larger cluster of advanced AI economies that includes Canada, and several clusters of emerging and nascent AI ecosystems. This structure suggests that the indicators capture meaningful similarities and differences in national AI strategies rather than random variation.



A simple linear regression model is then estimated, with the nine sub-scores as predictors of the overall score. Because all variables are on the same 0–100 scale and the sample contains 83 countries, a linear model is appropriate and easy to interpret. The fit is extremely strong ($R^2 \approx 0.9995$), indicating that the overall score can be almost perfectly reconstructed from a weighted combination of the underlying indicators. Using standardized predictors, the largest coefficients are found for commercial strength (about 3.05), research (2.70), development (1.93), and talent (1.89), followed by government strategy (1.45), infrastructure (1.19), and scale (1.04). Operating environment (0.59) and intensity (0.32) have smaller effects once other dimensions are controlled for.

Table Here

These results imply that the overall AI ranking is driven primarily by four forces. First, countries with strong commercial AI ecosystems—dense networks of AI firms, startups, and industrial adopters—score highest. Second, strong AI innovation capacity, reflected in research and development, is almost equally important. Third, human capital in the form of AI talent plays a major role. Fourth, the presence of a coherent government strategy and robust infrastructure helps translate research and talent into large-scale AI deployment. The predicted rankings match the actual rankings extremely closely, with a Spearman rank correlation of 0.9975 and almost identical positions for the top ten countries, which confirms that these drivers are embedded in how the index itself is constructed.

Part 2 — Strategy Proposal for Canada’s AI Development

We use standardized regression to identify the key capability dimensions that affect a country’s AI competitiveness. The size of each regression coefficient shows how much improving that capability can raise a country’s overall AI performance. A larger coefficient means that the capability explains more differences in AI rankings across countries and is therefore a strategically important area for improving national competitiveness. Based on our results, the dimensions with the strongest influence on the overall score are, Commercial (coefficient 3.046), Research (2.701), Development (1.930), Talent (1.892), Government Strategy (1.446), and Infrastructure (1.18).

After identifying these core dimensions, we examined Canada’s global standing through the Top 20 ranking chart. The results show that Canada performs relatively well in Commercial (6th) and Government Strategy (3rd), suggesting that Canada has advantages in AI commercialization and policy support. In contrast, Canada ranks lower in Research (10th), Talent (8th), and Infrastructure (18th). These weaker rankings appear in dimensions that have a large impact on overall competitiveness, which is why we consider Research, Talent, and Infrastructure to be the three areas Canada most urgently needs to strengthen.

These three dimensions also rely on and support each other. Computing power Infrastructure is the foundation. Without public GPU clusters and a unified compute platform, Canada cannot support large-scale AI research and development, and research teams will not be able to run

more complex experiments. If there are no strong models or research tools, Research will lack originality and depth, and scientific progress will be difficult to maintain. At the same time, without clear application needs, industry incentives, and improvements in the education system, the shortage of Talent will become more serious, and the current training system will not be able to meet the needs of research and industry.

Based on the Top 20 ranking chart, the United States, United Kingdom, France, and Germany are leading countries in Research, Talent, and Infrastructure. These countries perform much better than Canada in these important areas. Canada can learn from the policies that have worked well in these countries and use similar approaches to improve its own performance in these three core areas.

First, the United States uses a "foundation models + national compute platform" strategy that has really helped improve Research. America's AI Action Plan shows that the U.S. invests a lot of money in developing large models, helping companies like OpenAI, Google, and Meta train GPT, Llama, and Gemini. The government also funds basic science research through NSF and DARPA. At the same time, through NAIRR and Pillar II, the U.S. has built national compute systems including GPU/TPU supercomputers, data centers, power systems, and chip capabilities. They treat computing power as a national security resource and let universities and companies use it. Because they developed both models and computing power together, the U.S. has stayed ahead in Research for a long time.

Canada can learn from this by connecting the research work at Mila, Vector, and Amii with a single national compute platform to create a Canadian Foundation Model & Compute Platform. Using public GPU clusters and sharing computing resources can help fix Canada's Research problems that come from weak Infrastructure. This would create the same kind of "research-compute" positive cycle that the U.S. has.

Second, the model that the U.S. builds talent through a "national system for AI skills + tax incentives for corporate training" is also worth studying. America's AI Action Plan indicates that the U.S. enhances adult skills through vocational education and on the job training etc. Furthermore, the Treasury Department has issued guidance to clarify that AI training qualifies for tax-free reimbursement under Section 132 of the Internal Revenue Code which incentivizes businesses to invest.

Canada could adapt this model into a "Canadian AI Skills and Workforce Platform" that works across multiple government departments. This platform would be built on three core components: 1) A tax free reimbursement policy for corporate AI training. 2) A comprehensive AI literacy system integrated from basic education through higher education. 3) Practice-oriented training programs driven by industry-academia collaboration. This integrated approach would not just address the current talent shortage but also enhance the labour market's capacity to absorb and adapt the AI skills, thereby creating a positive cycle of "upskilling—industry application—policy optimization—value creation."

Also, the Infrastructure investments in the United Kingdom, France, and Germany show what's possible. The data and analysis show that Canada is at a similar level as these three countries, but they're investing much more in computing infrastructure. The UK put £900 million into building AIRR, including the Isambard-AI and Dawn supercomputing centers, with plans to increase computing power 20 times by 2030. France expanded the Jean Zay supercomputer through France 2030 and lets researchers and startups use it for free. Germany built JUPITER, Europe's first exascale supercomputer, and provides easy access to computing power for small and medium businesses through JAIF. Canada hasn't invested as much in national AI computing or made it as easy to access. If we don't move faster, the gap will probably get bigger in the next few years.

So Canada needs to take clear steps to improve Infrastructure: build public GPU clusters in Toronto, Montreal, and Edmonton, create a Compute Credits system for distributing access, and copy how the UK, France, and Germany make computing power available. This would make computing a public resource that both researchers and businesses can use.

Part 3 — Practical Steps for Implementing the Strategy

To achieve the above strategic goals, Canada needs to promote specific and executable actions from three levels: system, resource allocation and policy incentives. The following implementation steps fully correspond to the three major goals proposed in Part 2, forming a practical system that integrates Research, Talent and Infrastructure.

Strategy 1:

Establishing a National AI Compute and Model Coordination Body: Toward a Unified Governance Framework

According to the Pan-Canadian AI Strategy published by ISED and CIFAR, Canada's current AI research system is a "distributed research network" made up of Mila, Vector, and Amii, rather than a centralized national structure. Each of the three institutes operates its own independent GPU cluster. As a result, GPU and compute resources across the country are managed separately, with no unified planning or coordination. Public information also shows that Canada lacks large-scale compute capacity for training advanced AI models, and many research teams must rely on U.S. cloud services. Taken together, these facts show that Canada does not have a dedicated national body responsible for AI compute planning, public compute management, or foundation model development. This means the federal government cannot fully understand the country's GPU capacity, deployment layout, or infrastructure needs.

To address these structural problems, Canada needs to create a national coordination body for AI compute and model development. This body would not conduct research itself. Instead, it would take on national-level responsibilities such as planning, resource coordination, and public governance. It should be led by ISED, because ISED is the federal department responsible for innovation, science, and industrial development, and it has the authority to make technology

policy and coordinate resources across sectors. CIFAR should also be included, because it manages the Pan-Canadian AI Strategy and coordinates national investments and research networks. The three national AI institutes—Mila, Vector, and Amii—must participate as well, since they represent Canada's key AI research strength and have first-hand knowledge of compute needs, model development, and talent training. In addition, the NRC should be involved, as it is the national research organization responsible for technical evaluation and major scientific infrastructure.

The main role of this coordination body is to build a unified national plan for AI compute and foundation models. It would handle nationwide tasks such as forecasting GPU demand, planning where compute clusters should be placed, and coordinating the required supporting infrastructure. It would also set up management rules for public compute platforms, including how researchers apply for compute, who can access it, and what usage standards should be followed, to ensure fair distribution of resources. Finally, the body would define Canada's foundation model development direction and bring together related research funding and infrastructure budgets, allowing universities, research institutes, and companies to work within a unified national framework.

Strategy 2:

Building Canada's AI Supercomputing Centers: Creating an Open Public Compute Network

Building a supercomputing center is just the first step. Many countries have found that after building supercomputers, the usage rates stay really low. Either the application process is too complicated and researchers just decide to buy AWS instead, or the requirements are too high and PhD students and small companies can't access it. Canada has to avoid this trap. The solution is to learn from the UK's AIRR tiered access system[1]: PhD students use a "fast track" with 48-hour approval and get 500-2000 GPU hours per month to run experiments. Regular projects use a "standard track" where they submit a proposal for review and can get 10,000 to 100,000 GPU hours. National-level tasks (like training a Canadian Foundation Model) use a "priority track" with access to over 500,000 GPU hours. This way small projects don't get squeezed out by big ones, young researchers can go from "having an idea" to "starting experiments" in 2 days instead of several months, and major national projects get stable guaranteed access.

The second key point is to learn from France's Jean Zay "open science is free" policy[2]. The logic is simple: if taxpayer money trains the model, the results should benefit society. The rules work like this: publish your paper on arXiv, open-source your code on GitHub, upload your model to Hugging Face, and the computing is 100% free. If it's a commercial confidential project, you pay market price (but still 50% cheaper than AWS). For startups it's even more flexible: the first 6 months are completely free with no open-source requirements (giving time to explore the product). From months 6-18, if they promise to keep their IP in Canada and hire local employees, they keep getting a 50% discount. The goal of this policy is to turn public computing into an innovation accelerator, not an exclusive resource for a few big companies.

The third key point is to learn from Germany's JAIF "compute + technical support" one-stop service[3]. What happens if you get computing power but don't know how to use it? Many teams get GPU access and then realize they don't know how to write distributed training code or who to ask when they hit bugs. The solution is to have technical support teams of 15-20 people in Toronto, Montreal, and Edmonton (AI engineers, systems engineers, data engineers) whose job is to help users optimize code and fix problems. At the same time, develop an "AI Compute Toolkit" that provides pre-configured Docker images, training templates, and deployment tools that users can use right away. For small and medium businesses, offer special "compute vouchers + technical advisor" packages: each company gets \$50,000 CAD worth of free computing per year, plus an advisor who provides 20 hours of free consulting. Run regular "AI Compute Boot Camps" to teach PhD students and entrepreneurs hands-on how to use the system. In other words, the Compute Council shouldn't just build computing power—it needs to make sure that computing actually flows to where it's needed most: university labs, startups, and small businesses.

Although Canada has taken advantage in cultivating and attracting top AI research talent through institutions like CIFAR(1), the 44th place ranking in AI literacy highlighted in the KPMG(2025)(2) research report and the 30% vacancy rate for digital positions within government(3) indicate that this advantage has not effectively translated into widespread skill enhancement across the broader workforce. The core contradiction is the disconnection between research leadership and practical skills application in the labour market. The small and medium sized enterprises are unable to conduct effective training due to high costs.

In order to address this structural challenge and directly draw on the U.S. experience(AI training in the tax free reimbursement)(4), it is recommended that the Ministry of Finance spearhead the establishment of an "AI training expense reimbursement program". This program would allow enterprises to receive a certain amount of tax-free reimbursement after paying for employees' AI skills training, thereby directly reducing corporate training costs. Its scope would cover external training expenses paid by enterprises for their employees. At the same time, to provide targeted support for small and medium-sized enterprises, a separate "AI adoption grant" should be established to subsidize a portion of the costs associated with hiring certified AI consultants. Additionally, it should fund digital training initiatives for the public sector, thereby mobilizing all societal sectors to jointly participate in AI skills development and weaken the "job-skill mismatch" in the labor market.

Based on this foundation, the following three major strategies will be pushed simultaneously:

1. Strengthen and upgrade the Canadian government's existing strategic framework "Priority Three: Talent and Training" (3). The Treasury Board Secretariat will issue directives mandating foundational AI skills as compulsory annual training for all public servants and linking this to departmental performance evaluations to systematically address capability gaps in the public sector. Simultaneously, evolve the existing digital talent platforms from mere recruitment tools into a national level "AI Skilled Cloud

Platform". This integrated platform will combine skills assessment, personalized training pathways, and micro-credential certification."

2. Establish a comprehensive AI literacy education system spanning all stages of life. Led by the Ministry of Education, this initiative will extend AI literacy education beyond existing career and technical education and workforce training programs. General AI courses will be introduced in basic education(K-12) while higher education will develop modular, micro-certificate based AI application skill courses. This will position AI literacy as a core competency alongside numeracy and foster a sustainable talent pipeline.
3. Deepening the integration of AI with professional education. Through a joint initiative by the Ministry of Innovation, Science and Economic Development and the Ministry of Labour, the industry-led AI application courses will be fully integrated into academic systems across business, engineering, healthcare and other fields. The government matching funding will establish training centers and provide wage subsidies for co-op students. This can incentivize businesses to deeply engage in talent development directly.

Storytelling

Canada has a decent foundation in AI today, with three major research institutes, a relatively complete policy framework, and some talent reserves. However, Canada's weaknesses in computing power, research output, and talent supply are very obvious. This is especially true as international competition gets more intense - the United States, United Kingdom, France, and Germany are all speeding up their investments, which means the global gap keeps getting bigger.

To avoid falling further behind, we've built a clear strategy that starts with the most basic computing power and institutional setup, then moves to model development and talent system upgrades.

Stage 1 (Years 1-2): Establish National AI Coordination Mechanism

The first step is to create a national AI compute and model coordination body.

Its tasks include:

1. Making national compute plans, including GPU demand forecasts for the next few years and which cities should have major computing deployments.
2. Setting unified standards for building public GPU clusters so infrastructure in different regions can work together.
3. Getting the government, CIFAR, the three major AI institutes, and NRC to work together under one framework.

The focus of this stage is "organizing national resources." It will take about 1-2 years to set up and get into stable operation. We're using the Canada Infrastructure Bank as a reference because, like the proposed "national AI computing body," it's a federal Crown corporation

responsible for nationwide, cross-departmental, strategic infrastructure planning and investment, with similar governance complexity.

Stage 2 (Years 2-5): Build National Computing Infrastructure

Once we have unified planning, Canada can start building a real national computing platform.

Key tasks include:

- Build public GPU clusters in Toronto, Montreal, and Edmonton
- Deploy high-performance GPUs to form a nationwide shared computing network
- Set up access mechanisms (fast track, standard track, national track)
- Create technical support teams to help researchers and companies use the computing power

Stage 3 (Years 1-2): Launch Canadian Foundation Model (CFM)

Once the computing centers are built and running, Canada can start developing its own foundation models. This includes:

- Launch the first round of CFM training
- Develop both research and industry versions of models
- Set up a data sharing and open science framework
- Make it free for researchers and offer computing discounts to startups

Stage 4 (Years 2-5): Fully Build the AI Talent System

The talent system reform will move forward at the same time as computing and model development.

Key tasks include:

Implement tax-free reimbursement policy for AI training expenses: Let companies provide qualified AI training for employees and get tax-free treatment, with annual per-person limits set to directly lower company costs. At the same time, provide support through a matching "AI Application Subsidy" for SMEs hiring certified consultants.

Build a comprehensive AI literacy education system: Led by the Ministry of Education, introduce general AI courses in K-12 basic education, and develop modular, micro-credential AI application skills courses in higher education.

Deepen integration of AI with professional education: Fully integrate industry-led AI application courses into business, engineering, healthcare and other disciplines. Use government matching funds to set up training centers and provide wage subsidies for co-op students to companies.

Upgrade digital talent platforms: Evolve existing platforms from recruitment tools into a national-level AI Skills Cloud Platform that includes skills assessment, personalized training pathways, and micro-credential certification functions.

Strengthen public sector capabilities: Make basic AI skills mandatory annual training for all public servants and link it to performance evaluations.